

Supplementary Material

Hollingham (2026) — Newborough Warren Coastal Sand Dune Aquifer, Isle of Anglesey

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Supplementary Note S1: Interactive Map

Supplementary File S1: Interactive map of the full dipwell monitoring network (Google Maps), accessible at:

https://www.google.com/maps/d/viewer?mid=1hXLAauiMeaVsXhBR_IoUTziAtjk

Supplementary Note S2: Aquifer Geometry, Hydraulic Parameterisation and the Basis of the Spatial Flow Model

S2.1 The Borehole Data Problem

Darcy's law and any spatially explicit groundwater flow model require three spatially resolved inputs: hydraulic conductivity (K), aquifer thickness (b , from which transmissivity $T = K \cdot b$ is derived), and hydraulic gradient (∇h). For the Newborough Warren dune system, only the third of these — the hydraulic gradient — is well constrained, through the 88-dipwell monitoring network. Hydraulic conductivity and aquifer thickness are poorly constrained, because borehole logs providing lithological or geophysical evidence of the depth to the bedrock or glacial till substrate are absent across the great majority of the 700 ha dune system.

The available borehole evidence is confined to four locations documented in Betson et al. (2002), a report commissioned by the Countryside Council for Wales (Contract FC†73-05-18) covering hydrogeological investigations at the site. These boreholes provide minimum depth-to-basement constraints at four spatial points and constitute the only direct subsurface lithological evidence available for the site. They are summarised in Table S2.1.

Borehole	Easting (OSGB36)	Northing (OSGB36)	Sat. thickness (m)	Source
Water borehole	241,721	364,133	12.8 (min.)	Betson et al. (2002)
NH1	241,734	363,306	6.5	Betson et al. (2002)
NH2	242,024	363,107	6.5	Betson et al. (2002)
Borehole 3	242,837	363,288	3.65	Betson et al. (2002)

Table S2.1. Borehole constraints on saturated thickness from Betson et al. (2002). Values represent confirmed minimum depth to the low-permeability substrate (glacial till or bedrock). The Water borehole value of 12.8 m is a minimum — drilling did not reach the substrate. The eastern borehole (Borehole 3) shows progressive thinning toward the Menai Strait. These four boreholes constitute the only direct subsurface lithological evidence available for the 700 ha dune system and represent the most important data gap for any future spatially explicit groundwater model.

The tracer test reported by Betson et al. (2002) provides the single direct estimate of hydraulic conductivity for the site, yielding $K = 6.0$ m/day at the central dune plain (range 3.0–9.0 m/day from the sensitivity analysis). This value is used as the site-wide K constant where transmissivity is required for the Darcy flow direction vectors in the spatial figures (§4.9.6, Figure 57), with the range providing the basis for the exploratory uncertainty envelope noted in the figure annotations.

S2.2 The β_1 Proxy for Relative Aquifer Thickness

The physical interpretation of β_1 is the fraction of a unit rainfall pulse that reaches the water table as a head rise, integrated over the monthly timestep. In an unconfined aquifer, this is governed by the depth of the unsaturated zone and the specific yield: a thick unsaturated zone above a large storage volume attenuates the rainfall-to-head transfer, producing a low β_1 . A shallow, low-storage aquifer (thin unsaturated zone, low S_y) produces a high β_1 — rainfall arrives at the water table quickly and raises it sharply. Although β_1 cannot be inverted algebraically to yield aquifer thickness without independent knowledge of the recharge flux and specific yield, its spatial pattern is a reliable proxy for the relative contrast in aquifer thickness across the site.

The Eastern Block (C1: $\beta_1 = 4.576$; C2: $\beta_1 = 3.974$) and Western Residual cluster (C3: $\beta_1 = 3.573$) contrast is consistent with the geological interpretation of Stratford et al. (2007): the Eastern Block sits on shallow till and estuarine deposits, producing a thin, storage-limited aquifer with a flashy response; the Western Residual Cluster occupies deep, clean aeolian sand, producing a capacious, buffered aquifer with an attenuated response. The Forest cluster (C4: $\beta_1 = 2.487$) is anomalously low even relative to C3, reflecting canopy interception rather than additional aquifer depth — confirmed by the depth-dependent PET analysis (Script 15) which found $\lambda = 0$ optimal for C4, indicating that capillary disconnection at depth is not the primary control. C5 (Coastal Forest: $\beta_1 = 2.423$) shows a comparable value to C4, consistent with both clusters carrying the same Corsican pine canopy on the same deep sand substrate.

Cluster	Label	β_1	β_2	β_3	Geological context
C1	Lake Edge	4.576	0.956	0.088	Shallow till/estuarine substrate (NH1, NH2 boreholes: 6.5 m); rapid lake exchange
C2	Dune	3.974	1.762	0.064	Shallow till substrate consistent with C1; mature open dune
C3	Western Residual	3.573	1.833	0.057	Deep aeolian sand; Water borehole ≥ 12.8 m; DEM ridge geometry
C4	Main Forest	2.487	2.583	0.018	Same deep sand substrate as C3; low β_1 reflects 24% canopy interception
C5	Coastal Forest	2.423	1.307	0.044	Pine canopy on coastal sand; geomorphological thinning toward Menai Strait

Table S2.2. Cluster mechanistic coefficients (from Table 3, displacement-formulation SSM) and geological context. The β_1 contrast between Eastern Block (C1/C2) and Western Residual cluster (C3) is consistent with the stratigraphic interpretation of Stratford et al. (2007). C3 and C4 share the same deep aeolian sand body, confirmed by β_1 similarity after correcting for canopy interception and by Pearson affinity persistence post-felling. Borehole 3 (3.65 m, eastern margin) confirms progressive thinning toward the coast.

S2.2.1 Aquifer Thickness Surface (Developed for the PDE Model)

During development, a two-dimensional finite-difference partial differential equation (PDE) solver was implemented as an alternative spatial framework (see S3.3). That model required a spatially continuous aquifer thickness surface to compute transmissivity $T = K \cdot b$ at each grid cell. Because only four borehole constraints were available, an IDW-interpolated thickness surface was constructed using the borehole values as hard nodes supplemented by cluster-level thickness priors derived from the β_1 proxy argument above and per-well overrides at geological boundaries. The surface was interpolated on a 50 m grid using power-1 IDW with a physical minimum of 0.3 m enforced.

The PDE model was subsequently evaluated and rejected (S3.3) because it produced near-zero site-wide differences for forest management scenarios, primarily due to dilution of the modified β coefficients during grid interpolation, weak drainage feedback at C4, and the absence of ridge-derived lateral recharge from the source terms. The thickness surface is therefore not used in any published result. The per-well equilibrium framework adopted in the main paper (Section 3.8) operates directly from the fitted β coefficients at each well and does not require a thickness parameterisation. The Darcy flow direction vectors shown in Figure 57 are normalised head-gradient vectors derived from the interpolated mean head surface and are independent of aquifer thickness.

The borehole constraints in Table S2.1 and the geological contrasts summarised in Table S2.2 remain relevant for any future spatially explicit model. Slug tests at two to three representative wells per cluster and a ground-penetrating radar survey of aquifer thickness are identified as the highest-priority data gaps for future field campaigns (§5.9).

S2.2.2 Per-Well Geological Constraints

Several wells occupy geological settings that depart from their cluster's typical substrate. These are noted here for reference, as they would serve as boundary constraints for any future spatial model.

CEH14 (241,292 E, 364,488 N), classified as C4, sits at the crest of the bedrock ridge at 14.4 m AOD with a mean water table head of 13.3 m AOD. The water table is approximately 1.1 m below ground surface here. CEH14 overlies irregular bedrock topography on the elevated ridge flank, where buried ridges impede lateral drainage — consistent with its anomalous negative β_3 . That coefficient is not significant, however ($\beta_3 = -0.016$, $p = 0.24$), and six of the nine C4 wells likewise carry a non-significant β_3 , so CEH14 sits at the imprecise end of a cluster that is imprecise throughout; the negative sign should not be read as a result in its own right. Its water-balance residual is -0.011 m/month, the most negative in the network (§4.9.7, Figure 58).

Wells CEH7 (243,386 E, 363,613 N) and CEH8 (243,150 E, 363,382 N) sit at the far eastern estuarine margin of the site, where the DEM confirms ground elevation of 1.2–5.2 m AOD. Both represent coastal pinch-out positions where the aquifer thins to approximately 4.0 m. and their records are intermittent.

S2.3 Why C4 Retains Its Cluster Identity After Clearfell

A key question for the scenario analysis is whether the clearfell treatment wells (FE1, FE2, FE4, LIS1) converged toward C3 open-dune behaviour after the December 2017 felling. The Pearson affinity analysis (Section 4.3) provides a direct empirical test: all four impact wells maintained $r > 0.97$ affinity with C4 even in the post-felling period (2018–2026), with no significant shift toward C3 affinity.

This persistence is consistent with the interpretation that the Forest cluster signature at these wells reflects deep sandy substrate at high topographic elevation rather than canopy influence alone — in which case convergence toward open-dune behaviour may not be achievable regardless of management intervention. C4 wells sit at the highest topographic positions in the dune system (mean head 9.52 m AOD), and their deep unsaturated zones and low recharge sensitivity reflect their geological position, not only their canopy cover. This conclusion is independently supported by the NW10 broadleaf comparison (§5.6.3): NW10, situated within the 1993 clearfell and 1996 broadleaf restocking block, has maintained C4 cluster affinity ($r = 0.986$) throughout the 18-year monitoring period despite its different canopy cover, confirming that substrate position rather than canopy type is the dominant control on cluster identity.

S2.4 Reproducibility

The spatial model parameterisation documented in this note is implemented in 19_spatial_groundwater.py (scenario viewer data preparation, IDW thickness surface, per-well equilibrium calculations) and 20_spatial_figures.py (Darcy flow direction vectors, mean head surface, water balance residual field). The β_1 proxy analysis draws on the cluster-level SSM coefficients exported by 03_state_space_model.py (03_03_cluster_mechanistic_coefficients.csv). The Pearson affinity test for C4 cluster persistence post-felling is computed in 05_pearson_affinity.py. All scripts read intermediate data produced by the main pipeline and are maintained in the canonical pipeline sequence. The interactive scenario viewer (scenario_viewer.html) is a standalone browser application that reads the per-well coefficient and location CSVs exported by Script 19.

Supplementary Note S3: Scenario Modelling Framework and Limitations

S3.1 Purpose and Scope of the Scenario Analysis

An exploratory scenario analysis was developed to investigate whether the site's own SSM β coefficients could yield defensible quantitative predictions of water table response to management interventions and climate change. This note documents the scenario framework adopted (the per-well equilibrium framework described in Section 3.8), the alternative PDE-based approach that was evaluated and rejected during development, and the structural limitations that apply to any equilibrium representation of the system. The scenario outputs themselves are reported in Section 4.10 and discussed in Section 5.5.2 of the main paper; this note provides the technical reasoning behind the methodological choices and their limitations.

S3.2 The Per-Well Equilibrium Framework

The scenario framework adopted in this study is a well-level equilibrium calculation that applies scenario-specific perturbations to the SSM forcing terms (rainfall, PET, canopy interception, atmospheric draw) and computes the resulting change in equilibrium head for each reference well. Per-well Δh values are then interpolated to a 50 m grid on the British National Grid by Delaunay triangulation with linear barycentric weighting (`scipy.interpolate.griddata, method="linear"`), masked to the rectangular sea-boundary extent; the resulting field is rendered in the interactive scenario viewer rather than as a static map. The accompanying interactive scenario viewer (scenario_viewer.html) renders the same per-well Δh field using power-1 eight-nearest-neighbour inverse-distance weighting for responsiveness under slider-driven re-interpolation, and additionally offers a depth-below-

surface view (DEM minus interpolated head, per cell) with dune-ridge cells masked — see S3.7 for the viewer-specific rendering conventions. The full per-scenario × per-season × per-cluster Δh output is tabulated in the supplementary file 19_scenario_summary.csv.

The scenario head change for each well is computed directly from the fitted SSM equation as

$$\Delta h = (\beta_1 \cdot P_{eff,sc} - \beta_2 \cdot PET_{sc} - \beta_3 \cdot \bar{h}_{disp}) - (\beta_1 \cdot P_{eff,0} - \beta_2 \cdot PET_0 - \beta_3 \cdot \bar{h}_{disp})$$

where the $\beta_3 \cdot \bar{h}_{disp}$ term cancels algebraically between scenario and baseline expressions, so β_3 does not materially affect the computed Δh . Specific yield does not appear in the equation because the fitted β coefficients already embed the aquifer storage response through the SSM fit to observed head data; the per-well WTF-derived S_y values (Section 3.7.3) are retained as supplementary display information in the interactive viewer, not as a divisor on Δh . The framework is honest about what it is: a geometric aggregator of SSM responses across the cluster structure, not a physical model of groundwater flow.

The per-well framework has two important structural features that shape the interpretation of its outputs. First, scenario parameter changes for canopy interception and β_2 apply only to the C4 and C5 Forest cluster wells, because that is where the relevant physical changes occur under forest management. The framework therefore necessarily produces a response confined to the C4/C5 zone and its immediate interpolated surroundings, with effectively no propagation into C1, C2, or wider C3 clusters. Second, climate scenario parameter changes (ΔP , ΔPET) apply to all reference wells simultaneously, so the resulting Δh field is spatially continuous and reflects the site-wide β coefficient structure. The contrast between forest management and climate scenarios in the spatial outputs is therefore a direct consequence of which wells are affected by which perturbations, not an artefact of the numerical method.

S3.3 Evaluated-and-Rejected: The 2D Steady-State PDE Model

During development, an alternative spatial framework was implemented as a two-dimensional finite-difference PDE solver based on the Helmholtz form of the steady-state groundwater flow equation:

$$\nabla \cdot (T \cdot \nabla h) - \beta_3 \cdot h = -(\beta_1 \cdot P_{eff} - \beta_2 \cdot PET)$$

with transmissivity $T = K \cdot b$ ($K = 6.0$ m/day after Betson et al., 2002; b IDW-interpolated from four borehole constraints supplemented by cluster-level priors derived from the β_1 proxy, also after Betson et al., 2002). Dirichlet head conditions ($h = 0$ m AOD) were applied at sea boundaries, and implicit Neumann no-flow conditions at the ridge. The sparse linear system of approximately 4,200 equations was solved using `scipy.sparse.linalg.spsolve` (Virtanen et al., 2020).

This approach is well-posed mathematically — the Helmholtz structure guarantees a unique solution with the given boundary conditions — and was physically consistent in that the β coefficients entering the source term are the same parameters estimated from the observed well records. The solver produced physically plausible baseline head surfaces and Darcy flux fields qualitatively similar to those obtained from the per-well framework.

However, the PDE solver produced near-zero site-wide differences for forest management scenarios, for three structural reasons.

First, the forest management scenarios modify β coefficients only for the C4 and C5 Forest wells. These per-well β values are then IDW-interpolated to the 4,235-cell grid. Because the forest wells constitute a small minority of the 88-dipwell network, the modified β values are heavily diluted by the surrounding clusters during interpolation. The source term $\beta_1 \cdot P_{eff}$ –

$\beta_2 \cdot \text{PET}$ therefore changes by only a small amount at each cell, and the resulting change in the steady-state head solution is proportionally small.

Second, the internal drainage term $\beta_3 \cdot h$ provides a stabilising feedback in the Helmholtz equation. C4 has the lowest β_3 in the network (0.018, compared with 0.057 for C3 and 0.064 for C2; Table 3). This means that in the C4 zone the drainage feedback is weak, and a moderate source term change produces only a small equilibrium head change.

Third, and most fundamentally, the steady-state PDE does not represent the elevated boundary condition at the ridge margin. The observed mean head at C4 (9.52 m AOD) is substantially higher than the mean at C3 (6.09 m AOD), and this contrast is not a product of C4's β coefficients: the C4 wells occupy the highest topographic positions in the dune system, on and against the bedrock ridge flank. The PDE source terms are parameterised from the SSM β coefficients alone and carry no representation of that boundary, so the solution understates the C4 water table under both baseline and scenario conditions and the difference between the two solutions is correspondingly small. The argument rests on topographic position, which is directly observed. It does not rest on a ridge-derived lateral flux: the corrected water-balance residual field (§4.9.7, Figure 58) provides no evidence for one, since CEH14 — the well most proximal to the ridge — carries the most negative residual in the network (−0.011 m/month), the three wells above +0.02 m/month all sit in the open dune (NW2, T41a, NW1), and residual magnitude is uncorrelated with position on either axis (Spearman $\rho = -0.02$ on easting, $\rho = 0.91$; $\rho = -0.14$ on northing, $\rho = 0.27$). Whether any lateral flux exists at all cannot be resolved from the water-level record, and the rejection of the PDE approach does not depend on it.

A fully calibrated continuous-flow model would require, at minimum: (i) slug tests at two to three representative wells per cluster to constrain the spatial distribution of K (currently constrained only by the single tracer test of Betson et al., 2002); and (ii) a ground-penetrating radar survey of aquifer thickness, with coring, to replace the indirect β_1 -proxy constraints with direct measurements of thickness and specific yield. These are the same parameter-sparsity constraints that produced calibration difficulties in the Betson et al. (2002) MODFLOW model at this site.

For these reasons, the PDE approach was judged to overclaim relative to the data available and was rejected in favour of the per-well equilibrium framework. The framework represents what the SSM parameterisation supports without claiming spatial flow dynamics the monitoring data cannot constrain.

S3.4 Climate Scenarios in the Per-Well Framework

Climate scenarios applied under the per-well framework are based on the UKCP18 Regional 12 km projections for Wales (Met Office, 2018) and the CHES-SCAPE bias-corrected projections of Robinson et al. (2023). Two scenarios are carried forward into the Results (Section 4.10): UKCP18 2050s (2040–2069) applies a winter precipitation increase of +10%, summer precipitation decrease of −15%, winter PET increase of +5% and summer PET increase of +20%; UKCP18 2080s (2070–2099) applies +20%, −30%, +10% and +35% respectively for the same four terms. Both scenarios use RCP8.5 central-estimate (50th-percentile) values. Perturbations are applied per season, with winter defined as November–March and summer as May–September; the climatological baselines against which the perturbations are applied are the 2005–2026 monthly means from the RAF Valley climate record. The resulting Δh for each well is computed for each season separately, and the annual Δh is reported as the 0.5-weighted mean of the winter and summer equilibrium responses — this convention captures the physically significant seasonal asymmetry

between winter recharge and summer evaporative loss that an annual-mean forcing would obscure.

Three limitations on the interpretation of these outputs apply. First, the perturbations used are central-estimate values; the UKCP18 probabilistic range at the 10th and 90th percentiles spans roughly half to twice the central perturbation for precipitation and a broader range still for PET, so the Δh values reported here are a central point within a substantially wider envelope of plausible responses. Users of the interactive scenario viewer ([scenario_viewer.html](#)) can explore perturbations across the full 0.5–1.5× range of the seasonal sliders to visualise this envelope. Second, the equilibrium framework computes the steady-state response to sustained seasonal perturbations and does not resolve within-year dynamical propagation of recharge between winter and summer — for example, the effect of a wetter winter on the summer minimum through carry-over storage is captured only to the extent that the SSM β_3 drainage coefficient encodes it in the fitted mean. Third, the framework holds the SSM β coefficients themselves fixed at their historical values; it does not allow for structural changes in the β coefficients under future climate (for example, a higher β_2 under consistently warmer summers through increased vapour pressure deficit). Within these limitations, the climate scenario output is the most quantitatively defensible use of the equilibrium framework: every cluster's response is derived from its own fitted β coefficients under a uniform climate perturbation that reflects the current best estimate for the site's regional climate trajectory, and the result reflects the physical partition of rainfall and PET sensitivity that the SSM has estimated from 21 years of observation.

S3.5 Forest Management Scenarios in the Per-Well Framework

Forest management scenarios (full clearfell, 50% thinning, broadleaf conversion) apply canopy parameter changes only to the C4 and C5 Forest wells, and the framework correctly produces a response confined to the C4/C5 zone. This is a feature of the framework, not a bug: under the SSM parameterisation, canopy management directly affects only the wells beneath the canopy, and there is no structural mechanism for canopy effects to propagate into the open dune clusters beyond the hydraulic gradient already implicit in the baseline heads. The BACI monitoring record provides empirical corroboration of this structural feature: post-felling displacement is concentrated at the core impact wells, and wells outside the C4 cluster show no detectable felling-specific response once common-mode climate variability is accounted for (§4.6.4).

The clearfell scenario applies a β_2 multiplier of $\times 1.108$ at C4 and C5 wells, derived dynamically from the BACI-corrected Edge-tier post-felling β_2 ratio reported in §4.6.6 (Table 10): Edge mean ratio 1.101 minus Climate Control mean drift 0.993, plus 1.0. Canopy interception is set to 0%. The framework predicts modest annual water table rises at both forest clusters (C4: +0.028 m/month head, equivalent to +4.7 mm w.e./month; C5: +0.034 m/month head, +8.8 mm w.e./month), with a strongly asymmetric seasonal profile reflecting the dominance of summer PET in the β_2 sacrificial shielding mechanism established in §4.6.6 and §5.5.2.

The thinning scenario applies a β_2 multiplier of $\times 1.054$ (half the clearfell perturbation) with canopy interception reduced from 24% to 12%. The framework predicts approximately half the clearfell response at both forest clusters (C4: +0.014 m/month, +2.3 mm w.e./month; C5: +0.017 m/month, +4.4 mm w.e./month), as expected.

The broadleaf conversion scenario is represented by replacing the Corsican pine canopy interception fraction of 24% (Freeman, 2008) with an annual-mean deciduous interception of 15% (following the temperate deciduous meta-analysis of Komatsu et al., 2011) while holding annual-mean β_2 at baseline. No annual-mean justification exists for a year-round β_2

adjustment under broadleaf: the canopy is leafed in summer and leafless in winter, and the annual mean β_2 is expected to be comparable to pine. The framework predicts a near-neutral annual response at both forest clusters (C4: +0.010 m/month head; C5: +0.012 m/month head), masking a pronounced seasonal asymmetry: winter head-space values are the largest of any forestry scenario, but the summer response is negative at C4 (−0.007 m/month) as the growing-season transpiration penalty under mature deciduous canopy (β_2 summer multiplier $\times 1.113$) consumes the interception gain. The net annual volumetric effect is −3.4 mm w.e./month at C4 and +0.1 mm at C5. The seasonal asymmetry that underlies the phenological mechanism described in §5.5.2 — larger winter recharge pulse under leafless canopy, steepened hydraulic gradient, accelerated drainage into summer — is a dynamical trajectory response that the equilibrium framework does not resolve.

For the Figure 71 hydrograph (§5.7.4), a seasonally-varying β_2 is applied to the broadleaf scenario to represent the full-LAI summer canopy and leafless winter canopy. This seasonal parameterisation is a visualisation device and is not part of the §4.10 Results; it is used specifically to communicate the phenological argument from §5.5.2 alongside the BACI-observed record. The annual-mean output in 19_scenario_summary.csv remains the quantitatively citeable framework result.

S3.6 Implications and Future Work

The per-well equilibrium framework is a lightweight analytical device for translating the SSM β coefficients into spatial summaries of equilibrium head response. Its strengths are that it makes no claims beyond what the SSM itself supports, that it is fully reproducible from the published CSV outputs, and that it is explorable through an interactive viewer that exposes all parameter choices to the user. Its limitations are those of any equilibrium framework: it does not resolve dynamical trajectory responses, it does not capture cluster-to-cluster hydraulic coupling, and it cannot simulate transient interventions in which the time evolution of the water table matters for ecological outcome. The framework is adequate for the questions the paper addresses — whether forest management in the C4/C5 zone affects summer minima in C1 and C2, and how climate perturbations partition across the cluster structure — and the BACI monitoring record confirms the framework's structural conclusions.

Answering a different class of question, such as how a hypothetical lateral flux along the CEH14 ridge margin would propagate through the aquifer, would require a calibrated continuous-flow model. For that, the critical unknowns are hydraulic conductivity (currently constrained by a single tracer test at $K = 6.0$ m/day; Betson et al., 2002) and aquifer thickness (inferred from four borehole logs across 700 ha). Slug tests at two to three representative wells per cluster and a ground-penetrating radar survey of aquifer thickness are identified as the priority field measurements required to enable a properly calibrated continuous-flow model, should future work pursue intervention scenarios where lateral hydraulic propagation is central to the management question.

Summary of scenario modelling approach

Primary framework: Per-well equilibrium with triangulation-linear grid rendering (Sections 3.8, 4.10; scenario_viewer.html; 19_scenario_summary.csv)

Evaluated and rejected: 2D steady-state Helmholtz PDE solver (Section 3.8 PDE paragraph; S3.3 above)

Visualisation device: Figure 71 synthetic hydrograph with seasonally-varying broadleaf β_2 (§5.7.4; Script 21)

Empirical corroboration: BACI monitoring record (§4.6)

S3.7 Interactive Viewer: Rendering Conventions and Depth-Mode Visualisation

The companion interactive scenario viewer ([scenario_viewer.html](#); Hollingham, 2026b) renders the per-well Δh field described in S3.2, but uses a distinct grid-rendering method at the browser to preserve responsiveness under slider-driven re-interpolation. The viewer implements inverse-distance weighting with power 1 and eight nearest neighbours (10 m distance floor; 5 m exact-value shortcut) rather than the triangulation-linear method used for the static figures. Power-1 IDW with a fixed neighbour set produces a smoother surface than the naïve power-2 IDW previously used, removes the bullseye halo artefacts that arose around individual wells on the Δh surface, and is sufficiently close to the triangulation-linear rendering at well locations that the two methods agree to within the mapped colour-scale resolution across the interpolated interior. At well locations the viewer returns the well's observed value unchanged; outside the convex hull of the wells it extrapolates under the IDW kernel where the triangulation-linear method would return no data. This distinction is most visible near the south-eastern coastal margin, where the viewer renders a short extrapolated zone that the static figures correctly leave blank.

The viewer offers three complementary display modes. The first two — change in water table elevation (Δh) and absolute water table elevation (m AOD) — are elevation-based surfaces, and are the two quantities exported to `19_scenario_summary.csv`. The third mode, depth below surface (m), is a viewer-specific visualisation derived in the browser. It is computed per grid cell as $d = h_DEM(E, N) - h_WT(E, N)$, where h_DEM is the bilinearly-interpolated LiDAR DEM elevation at cell centre and h_WT is the IDW-interpolated scenario water table elevation at the same cell. This per-cell subtraction is the quantity that mediates the ecological response: the Curreli et al. (2013) slack vegetation thresholds are defined in depth-below-surface terms (SD15b wet slack summer viability limit: 0.61 m; SD16 dry slack viability limit: 0.98 m), and rendering the scenario water table against the topographic surface permits direct visual identification of the slack positions where scenarios cross these thresholds. The viewer's depth-mode colour scale is anchored on these thresholds: light blue at the 0.10 m winter flooding limit, green at 0.61 m, orange at 0.98 m, and deep red at 1.50 m, with explicit black tick marks on the legend at the SD15b and SD16 threshold depths.

A dune-ridge mask is applied exclusively in depth-below-surface mode. The rationale is that the water table is a continuous surface in space, and its elevation relative to ordnance datum is well-defined at every point within the site regardless of whether the overlying ground is a ridge or a slack floor; elevation-based maps (modes one and two) therefore require no topographic masking. Depth below surface, however, is meaningful only where the ground is approximately at the well-plane elevation — at a dune ridge the interpolated water table sits several metres below ground, but this is a consequence of the ridge height, not of any hydrological feature, and rendering such cells would produce a misleading strong-red patch on the depth map that obscures the ecologically relevant slack-floor response. The viewer masks a cell in depth mode when the DEM elevation at that cell exceeds the IDW-interpolated well-plane elevation (computed by the same IDW kernel applied to well DEM ground elevations) by more than 1.0 m. This threshold is user-toggleable from the viewer's control panel; disabling it reveals the unmasked depth field, which can be useful for diagnostic inspection but should not be used for ecological interpretation.

Additional viewer conventions of note: the basemap is a LiDAR-derived greyscale hillshade rendered at approximately 35% opacity underneath the scenario surface, matching the static figure hillshade convention of `map_utils.load_dem_hillshade` with azimuth 315°, altitude 35° and vertical exaggeration 3; climate perturbations can be explored across the full 0.5–1.5× range per season, extending substantially beyond the UKCP18 2050s and 2080s central-estimate presets to allow sensitivity analysis across the probabilistic envelope; and tooltips on any well display both the baseline and scenario-perturbed depth values alongside the

absolute head and Δh , providing a direct visual link between the per-well framework output and the ecological depth-below-surface interpretation.

S3.8 Reproducibility

The per-well equilibrium framework and climate/forestry scenario calculations are implemented in `19_spatial_groundwater.py`, which exports the full per-scenario × per-season × per-cluster output to `19_scenario_summary.csv` and generates the interactive scenario viewer. The interactive scenario viewer (`scenario_viewer.html`) implements the browser-side IDW rendering described in S3.7 and reads the same per-well data. The synthetic mean-year hydrograph (Figure 71) is generated by `21_forestry_scenarios.py`, which reads BACI coefficients from the Script 10 clearfell suite outputs. All scenario parameters — UKCP18 perturbation fractions, canopy interception values, β_2 multipliers, and seasonal definitions — are defined as named constants at the head of each script and are fully documented in the `scenario_viewer.html` source.

Supplementary Note S4: Specific Yield Mapping

S4.1 Per-Well Estimates

Table S4.1. Individual Well WTF Specific Yield Estimates — Newborough Warren 2005–2026: event-based median specific yield (Sy) derived from the water table fluctuation (WTF) method (Healy and Cook, 2002) for the 66 reference-network wells. Cl. = cluster under the k = 5 partition (C1 Lake Edge, C2 Dune, C3 Western Residual, C4 Main Forest, C5 Coastal Forest). n = number of qualifying monthly rising-limb events (criteria: $\Delta h > 5$ mm, net recharge $P - PET > 10$ mm). Q25/Q75 = interquartile range; a wide IQR reflects month-to-month variability in event estimates rather than measurement error. Wells in the forest clusters C4 and C5, marked Int. corr. = Yes, have net recharge adjusted for 24% canopy interception (Freeman, 2008). CEH12 (bedrock ridge — WTF response reflects fractured rock) and CEH15 (forest slack floor — slack topography dominates water table dynamics) are excluded from the IDW interpolation surface and do not appear here. Source: 18_wtf_01_well_sy_estimates.csv.

Well	Cl.	n	Sy med	Sy Q25	Sy Q75	Int. corr.
CEH11	C1	59	0.235	0.158	0.307	
CEH23	C1	44	0.207	0.123	0.252	
CEH25	C1	54	0.178	0.134	0.246	
CEH26	C1	49	0.214	0.115	0.276	
CEH27	C1	52	0.216	0.140	0.298	
CEH5	C1	65	0.194	0.159	0.275	
CEH6	C1	70	0.216	0.123	0.347	
CEH10	C2	64	0.263	0.161	0.353	
CEH24	C2	52	0.239	0.168	0.317	
CEH28	C2	41	0.236	0.183	0.314	
D10	C2	50	0.279	0.222	0.393	
D15	C2	54	0.288	0.232	0.370	
D17	C2	54	0.271	0.198	0.351	
D38	C2	54	0.227	0.172	0.287	
D41	C2	46	0.240	0.191	0.312	
D43	C2	59	0.271	0.212	0.347	
D44	C2	50	0.220	0.164	0.316	
D5	C2	49	0.274	0.202	0.348	
D6	C2	57	0.267	0.202	0.360	
D7	C2	50	0.286	0.220	0.323	
D8	C2	47	0.247	0.179	0.371	
D9	C2	49	0.259	0.203	0.362	
L7	C2	63	0.286	0.223	0.337	
NW3	C2	54	0.267	0.190	0.354	
NW4	C2	58	0.265	0.173	0.339	
NW4B	C2	54	0.240	0.159	0.318	
T41A	C2	55	0.227	0.172	0.323	
T41B	C2	45	0.215	0.159	0.313	
T41C	C2	47	0.215	0.189	0.308	
T41D	C2	47	0.254	0.197	0.334	
WMC1	C2	51	0.274	0.207	0.406	
CEH1	C3	67	0.273	0.226	0.346	
CEH18	C3	53	0.364	0.284	0.433	
CEH21	C3	31	0.413	0.356	0.445	
CEH36	C3	38	0.349	0.297	0.422	
CEH39	C3	38	0.349	0.287	0.407	
CEH4	C3	53	0.349	0.255	0.400	
CEH40	C3	27	0.285	0.243	0.323	

Well	Cl.	n	Sy med	Sy Q25	Sy Q75	Int. corr.
CEH41	C3	36	0.306	0.256	0.345	
CEH42	C3	34	0.351	0.284	0.389	
CEH9	C3	53	0.347	0.275	0.395	
D25	C3	45	0.289	0.242	0.393	
NW1	C3	72	0.259	0.215	0.369	
NW11	C3	68	0.285	0.229	0.336	
NW13	C3	50	0.249	0.206	0.334	
NW2	C3	68	0.252	0.212	0.342	
NW5	C3	56	0.305	0.250	0.350	
NW6	C3	45	0.281	0.206	0.346	
NW7	C3	47	0.333	0.280	0.398	
WMC2	C3	53	0.343	0.295	0.412	
WMC3	C3	36	0.328	0.279	0.423	
WMC4	C3	51	0.250	0.208	0.353	
CEH13	C4	50	0.231	0.172	0.325	Yes
CEH14	C4	55	0.194	0.126	0.265	Yes
CEH2	C4	60	0.241	0.188	0.337	Yes
CEH20	C4	50	0.254	0.208	0.312	Yes
CEH30	C4	48	0.278	0.211	0.355	Yes
CEH32	C4	49	0.240	0.181	0.366	Yes
CEH33	C4	48	0.255	0.201	0.321	Yes
CEH34	C4	50	0.264	0.196	0.342	Yes
NW10	C4	68	0.256	0.199	0.331	Yes
CEH16	C5	51	0.284	0.237	0.398	Yes
CEH17	C5	40	0.322	0.248	0.406	Yes
CEH19	C5	40	0.337	0.278	0.422	Yes
CEH31	C5	42	0.295	0.235	0.387	Yes
NW9	C5	60	0.308	0.241	0.382	Yes

S4.2 Reproducibility

The WTF specific yield estimates tabulated in Table S4.1 are computed by `17_wtf_specific_yield.py`, which identifies qualifying monthly rising-limb events, applies the canopy interception correction for C4 wells, and exports per-well Sy statistics to `17_wtf_well_sy.csv`. The spatial IDW interpolation of Sy values and the exclusion of CEH12 and CEH15 from the interpolation surface are implemented in `18_wtf_spatial.py`, which generates the Sy map figure and exports the interpolated surface. Both scripts read intermediate data from the main pipeline (`01_wells_clean.csv`, `01_climate.csv`, `01_locations.csv`, `02_cluster_stats.csv`) and are maintained in the canonical pipeline sequence.

Supplementary Note S5: Residual-Lag Test for Ridge-Derived Recharge

S5.1 Purpose

An earlier boundary-subsidy argument attributed a persistent water-balance residual at forest-margin wells — most visibly a positive residual at CEH14 — to lateral recharge derived from the northern rock ridge. It rested on two lines of evidence: a spatially structured pattern of positive residuals concentrated along the forest–dune boundary, and a closure argument that no other plausible flux could account for the monthly deficit. The corrected residual field (Script 20, 2026-08-06; §4.9.7, Figure 58) removes both. Sixty of the sixty-six reference wells now fall within ± 0.01 m/month, the field shows no gradient on either axis, the three wells above $+0.02$ m/month all sit in the open dune, and CEH14 is the most negative well in the network at -0.011 m/month. The balance closes without requiring an additional flux. The residual-lag analysis described below was designed as an independent, falsifiable test of the transport mechanism and is retained as a bound on what the 21-year monthly record could have detected: if ridge-derived recharge is delivered as a time-varying pulse, then water travelling from the ridge to successive dune wells must arrive at those wells with travel times that increase as a function of distance. The time structure of the SSM residuals, cross-correlated against rainfall, should therefore carry a distance-dependent lag signature. This Note documents the test and reports the null result.

S5.2 Extended State-Space Model

The single-period SSM fitted in Section 3.4.3 expresses the water table increment as a function of contemporaneous rainfall, where $h_disp_prev(t) = \text{DRAINAGE_DATUM} + h(t-1)$ is the displacement above the 3.7 m drainage datum at the start of month t :

$$\Delta h(t) = \alpha + \beta_1 \cdot P(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot h_disp_prev(t) + \varepsilon(t)$$

A preliminary analysis (Script 22) demonstrated that the residuals $\varepsilon(t)$ from this formulation carry a generic lag-1 rainfall signal at every well on the site, irrespective of cluster or position. This is the expected monthly-timestep consequence of the vadose zone: recharge takes approximately one month to propagate to the water table, so monthly-averaged water-table response lags monthly-averaged rainfall by roughly one month. This generic signal dominates the cross-correlation function at every well and masks any ridge-specific structure that might be present at longer lags.

To remove this confound, the residual-lag test uses an extended model that explicitly includes both $P(t)$ and $P(t-1)$ as regressors:

$$\Delta h(t) = \alpha + \beta_{10} \cdot P(t) + \beta_{11} \cdot P(t-1) - \beta_2 \cdot PET(t) - \beta_3 \cdot h_disp_prev(t) + \varepsilon'(t)$$

The β_{10} and β_{11} coefficients absorb the vadose-zone response between them; any remaining lag structure in $\varepsilon'(t)$ is therefore post-vadose and is the candidate ridge-transport signal. Fitted coefficients show that the extended parameterisation is only partly identified: β_{11} is statistically significant ($p < 0.05$) at 27 of the 64 wells, and its sign is mixed (negative at 33 wells, positive at 31), so the two rainfall terms do not partition the vadose-zone response cleanly at every well. The cluster-median lagged-response fraction $\beta_{11} / (\beta_{10} + \beta_{11})$ nonetheless rises across the open-dune-to-forest sequence, from -0.08 in C1 and -0.03 in C2 to $+0.11$ in C3 and $+0.28$ in C4 (C5 $+0.16$), consistent with the slower drainage of forest soils. Mean R^2 rises from 0.72 under the single-period model to 0.75 under the extended model across the same 64 wells. The extended model is used solely for this diagnostic test; the β_1 , β_2 , β_3 and α values reported in the main text remain the authoritative parameterisation for all other analyses.

S5.3 Cross-Correlation and Pre-Whitening

For each well with at least 140 months of data ($n = 64$ after excluding CEH3, CEH4, CEH7, CEH8 and CEH37 for reasons noted below), the Pearson cross-correlation $r(\epsilon'(t), P(t - N))$ was computed for lags $N = 0$ to 12 months. Prior to cross-correlation, both the rainfall and residual series were pre-whitened using a first-order autoregressive filter: $x'(t) = x(t) - \phi \cdot x(t-1)$, where ϕ is the AR(1) coefficient of the rainfall series (+0.211 at RAF Valley). Wells whose residuals retained AR(1) structure above $|\phi| = 0.2$ were additionally pre-whitened against their own residual autocorrelation before the common filter was applied. Statistical significance was evaluated using the Bartlett 95% confidence interval for a white-noise series, $|r| \geq 1.96/\sqrt{N}$, which for typical series lengths gives a threshold of approximately 0.15.

Wells excluded from the test: CEH3 (tidal boundary — outside the SSM operational domain), CEH4 (coastal erosion drift plus post-2017 clearfell drawdown confounding any lag signal), CEH7, CEH8, CEH37 (standard upstream exclusions carried over from §4.9.4). Wells beyond the 140-month minimum were retained without further filtering.

S5.4 Hypothesis Test

The test metric is the Spearman rank correlation between each well's peak-correlation lag N^* and its Euclidean distance from the ridge reference point at ($E = 241750$, $N = 364500$, OSGB36) — a representative coordinate for the nearest point of the northern rock ridge to the dune field. Under the ridge-transport hypothesis, Spearman ρ should be significantly positive.

Of 64 wells fitted, 51 had a peak cross-correlation that exceeded the Bartlett significance threshold. Across these 51 wells the peak lag has a mean of 2.67 months, a median of 2 months and a standard deviation of 2.29 months: 43 wells peak at lag 2, five at lag 3, and three at lags of 11 or 12 months with no systematic geographic pattern. The Spearman rank correlation between peak lag and ridge distance was $\rho = +0.002$ ($p = 0.99$) — no distance structure whatever. Cluster mean peak lags are 2.00 months at C1 and C2, 3.00 at C3 and 3.67 at C4 (C5 has a single significant well, at lag 11). A Mann-Whitney comparison of the C4 ridge-adjacent wells against the C2 and C3 dune-body wells does return significantly longer lags at C4 (3.67 against 2.31 months; $U = 233.0$, $p < 0.005$), but that ordering is a cluster-level contrast in drainage behaviour rather than a distance effect: within the network as a whole, lag is uncorrelated with ridge distance. The hypothesis of distance-dependent ridge transport is not supported.

The magnitude of coupling does not single out the ridge margin. Mean peak $|r|$ among significant wells is highest at C1 (0.274) and C2 (0.224), and the ridge-adjacent C4 wells (0.179) are indistinguishable from the geologically separate C3 wells (0.181), with C5 lowest (0.138). At the individual wells previously highlighted, CEH14 peaks at $r = +0.21$, CEH34 at +0.20, CEH2 at +0.18 and CEH13 at +0.16, against NW10 at +0.19 and NW1 at +0.13 in C3 — a spread too narrow to support a ridge-specific coupling signal. Neither the strength of rainfall coupling nor its timing is distance-structured.

S5.5 Interpretation

The null distance-lag result does not rule out ridge-derived recharge. It rules out one specific form of it: event-driven, pulse-delivered transport in which monthly rainfall on the ridge propagates to the dune field with travel times that scale with distance. The alternative interpretation — that ridge recharge is delivered as a near-steady baseflow, sufficiently smoothed by its transit through fractured bedrock that its month-to-month variance falls

below the detection threshold of monthly water-level records — is consistent with the null. A steady baseflow of this kind would manifest observationally as a constant positive contribution to the local water balance at wells along the flow path, which is precisely what the SSM residual α already absorbs. That predicted signature is, however, absent from the corrected residual field: the C4 median is -0.005 m/month and no forest well carries a residual above $+0.02$ m/month. The steady-baseflow reading is therefore consistent with the lag-test null but unsupported by the water balance, and the two mechanisms remain indistinguishable from water-level data alone only in the weak sense that neither is evidenced.

The main report's treatment of the water-balance residual (§5.2.1) no longer advances a ridge contribution: with the corrected field the balance closes without requiring an additional flux, and the residual is discussed there as a model-adequacy diagnostic rather than as evidence of a boundary subsidy. Definitive resolution of the transport mechanism would require either a ridge-crest rain gauge paired with bedrock piezometers, or direct geochemical tracer work on the ridge–dune flow path. Both are identified as priority further work in §5.9.

S5.6 Reproducibility

The residual-lag test is implemented in two standalone analysis scripts, `22_residual_lag_analysis.py` and `23_ridge_recharge_lag_test.py`, which are maintained as supplementary diagnostics (Phase 12) alongside the main pipeline to preserve the reproducibility of the headline results from the canonical pipeline sequence. Script 22 generates the residuals and AR(1) diagnostics; Script 23 fits the extended model, computes the cross-correlations, and applies the Spearman test. Both scripts read the same intermediate data produced by the main pipeline (`01_wells_clean.csv`, `01_climate.csv`, `01_locations.csv`, `02_cluster_stats.csv`) and produce independent outputs under `22_residual_lag_analysis/` and `23_ridge_recharge_lag_test/` in the project outputs tree. A plain-text summary of the hypothesis test result is written to `23_05_hypothesis_test_summary.txt` at runtime.

Supplementary Note S6: Seasonal Climatology Diagnostic of the Residual

S6.1 Purpose

Supplementary Note S5 reported a null result for event-driven ridge transport. This Note applies the same logic to a second candidate mechanism that was advanced when the water-balance residual was believed to carry spatial structure: that the Thornthwaite PET estimate underestimates the true summer atmospheric demand on the water table because it is temperature-only and does not capture net radiation, vapour pressure deficit, wind, or surface-condition effects. As with S5, the corrected residual field leaves no anomaly for this mechanism to explain, and the diagnostic is retained for what it establishes about the monthly time-series residual $\varepsilon(t)$ rather than as a competing explanation. Thornthwaite PET is an empirical index of atmospheric evaporative demand derived from temperature and day length; it does not define a specific reference surface. If that were the dominant explanation, the residuals should carry a systematic summer-negative signature that the fitted β_2 has not absorbed. This Note tests for that signature, along with two subsidiary diagnostics that bear on the remaining candidate explanations (steady ridge baseflow, nonlinear recharge response, and residual model error). The residual quantity throughout S5 and S6 is the monthly time-series SSM residual $\varepsilon(t)$, which is distinct from Script 20's scalar per-well water-balance closure residual discussed above and is unaffected by the 2026-08-06 correction to the latter; only the framing of these two Notes changes.

S6.2 Diagnostics

Three complementary tests are computed from the residuals $\varepsilon(t)$ produced by Script 22 for the same 64 wells that passed the ≥ 140 -month record-length filter and the site-extent exclusions detailed in S5.3.

(i) Seasonal climatology. For each well, the monthly climatology of the residual is computed as the mean of $\varepsilon(t)$ by calendar month over the record, giving a 12-point annual cycle. A sinusoidal fit $\varepsilon(m) = a_0 + a_1 \cdot \cos(2\pi m/12) + a_2 \cdot \sin(2\pi m/12)$ yields an amplitude $\sqrt{a_1^2 + a_2^2}$ and a phase (month of peak). Summer-minus-winter magnitude (JJA mean – DJF mean) is reported as a robust non-parametric proxy for the same quantity.

(ii) Independent ET proxy. A correlation of the residual against PET itself would be trivially zero because OLS fitting makes residuals orthogonal to every regressor by construction. Instead, the residual is correlated against monthly sunshine hours from the RAF Valley record — a direct radiation-based measurement that is not in the regression. If Thornthwaite underestimates summer ET in a way that β_2 has not absorbed, high-insolation months should carry disproportionately negative residuals, yielding a systematic negative Pearson r .

(iii) Spatial pattern within clusters. For C3 (Western-block open dune), the cluster is split by Euclidean distance from the ridge reference point used in S5 at a 1 km threshold (5 forest-adjacent wells, 13 warren-interior wells), and a Mann-Whitney test asks whether forest-adjacent wells have systematically smaller seasonal amplitudes than warren-interior wells — as would be expected if a steady ridge baseflow were flattening the residual at ridge-proximal locations.

S6.3 Result: the ET Hypothesis Is Not Supported

The sunshine-hours correlation is consistently negative in sign across every cluster, with network mean $r = -0.026$ and per-cluster means ranging from -0.001 (C4 Main Forest) to

−0.058 (C1 Lake Edge). None of the 63 wells exceeds the Bartlett 95% significance threshold of $|r| = 0.15$ — all 63 fall inside the null band. The sign is directionally consistent with Thornthwaite slightly underestimating radiation-driven ET, but the magnitude of the bias is below the resolution of the monthly water-level data to detect.

Summer-minus-winter residuals are negative in every cluster except C5, where the sign reverses to +0.003 m, with magnitudes of order −0.004 m (C2) to −0.008 m (C3). These values are an order of magnitude smaller than the month-to-month variability of the residuals themselves (typical well-level residual standard deviation of order 0.1 m) and should not be read as a summer-ET signal. Notably, C4 — the cluster where canopy interception would most plausibly produce Thornthwaite miscalibration — has the weakest sunshine correlation of any cluster, indicating that the cluster-mean β_2 has absorbed forest-specific ET behaviour reasonably well despite the simplicity of the fitted model.

The C3 within-cluster split also returns a null: forest-adjacent C3 wells ($n = 7$) have marginally smaller seasonal amplitudes than warren-interior wells ($n = 10$) — 0.0099 m against 0.0101 m; Mann-Whitney $U = 33.0$, $p = 0.44$, alternative “adjacent < interior”. The direction is the one a steady-ridge-baseflow mechanism would predict, but the difference is a fifth of a millimetre and nowhere near significant, so it carries no evidential weight either way.

S6.4 What the Residuals Actually Look Like

The per-cluster seasonal climatologies reveal a structure that fits none of the candidate mechanisms cleanly: the annual cycle is bimodal, with positive residuals in January–February AND June–July, and negative troughs in April–May and September–October. The sinusoidal fit captures this poorly, which is why the fitted amplitudes appear modest relative to the visible annual range. The fitted sinusoid peaks in late autumn or winter in every open-dune cluster — C1, C2 and C3 all in December, C5 in November — and later, in February, at C4, the shift at the forest cluster being consistent with its slower drainage. Across the network, 48 of the 64 wells peak between November and March and not one peaks in the May–August summer band.

This bimodal structure is not the signature of unmodelled summer ET, which would produce a single summer trough. It is not the signature of flat steady ridge baseflow, which would produce no annual cycle at all. And it is not a simple winter-recharge-nonlinearity signature, which would produce a single winter peak. Its most parsimonious interpretation is that a linear lumped-parameter model with time-invariant β coefficients is a rough approximation of a system whose soil-moisture storage operates nonlinearly across the annual cycle — wet-winter saturation increasing recharge efficiency beyond what the cluster-mean β_1 represents, shoulder-season soil drying reducing it below, and the whole cycle modulated by seasonal variation in vegetation water use that a constant β_2 cannot track.

The spatial pattern of seasonal amplitude nonetheless shows a coherent structure: the two largest amplitudes in the network are CEH14 (0.040 m) and CEH13 (0.026 m), both C4 forest-margin wells, against a network median of order 0.010 m; C4 has the largest cluster-mean amplitude (0.015 m) of any cluster. The two fields do not coincide, however: CEH14 carries the most negative water-balance residual in the network (−0.011 m/month; Figure 58, §4.9.7), at the opposite end of that distribution from the three open-dune wells that lead it. The rest of the C4 cluster sits within the general network spread rather than standing out. The elevated seasonal variance at CEH14 and CEH13 is therefore not accompanied by an elevated water-balance residual, which weakens the reading that larger seasonal boundary fluxes are responsible and leaves the alternative — that cluster-mean β values are a worse approximation at ridge-margin wells than at dune-interior wells — as the better-supported of the two. Water-level data alone cannot settle it.

S6.5 Combined Conclusion from S5 and S6

Taken together, Supplementary Notes S5 and S6 have excluded the two leading specific attributions of the water-balance residual:

The residual does not show the distance-dependent lag signature of event-driven ridge transport (S5).

The residual does not show the summer-negative signature of unmodelled Thornthwaite underestimation of atmospheric demand (S6).

The main text of §5.2.1 has been framed accordingly: the model explains 50–70% of month-to-month variance in Δh depending on cluster, and the unexplained portion is real and spatially structured (concentrated at the forest margin) but cannot be uniquely attributed to any single physical mechanism from water-level data alone. The fitted residual α remains the most defensible summary statistic for the steady-state boundary contribution; the underlying mechanism is most plausibly a combination of steady ridge baseflow, nonlinear soil-moisture storage dynamics not captured by the linear SSM, and minor β coefficient miscalibration, in proportions that cannot be resolved without additional measurement of the kind identified as priority further work in §5.9.

S6.6 Reproducibility

This analysis is implemented in `24_residual_seasonality.py`, which like Scripts 22 and 23 is maintained as a supplementary diagnostic (Phase 12) alongside the main pipeline to preserve the reproducibility of the headline results. The script reads the same intermediate data as Script 22 (`01_wells_clean.csv`, `01_climate.csv`, `01_locations.csv`, `02_cluster_stats.csv`) plus the raw RAF Valley climate file for sunshine hours (`RAF_Valley_Climate.csv`). All outputs are written to `24_residual_seasonality/` in the project outputs tree: a per-well climatology table (`24_residual_climatology.csv`), the per-cluster climatology panels (`24_01_climatology_panels_by_cluster.png`), the spatial amplitude map (`24_02_seasonal_amplitude_map.png`), the sunshine-correlation scatter (`24_03_sun_residual_correlation.png`), the per-cluster phase distribution (`24_04_phase_by_cluster.png`), and a plain-text summary of the diagnostic result (`24_05_diagnostic_summary.txt`).

Supplementary Note S7: Equilibrium Wetness Index — Per-Well Reconstruction

S7.1 Per-Well Reconstruction

The equilibrium wetness index (EWI) is the steady-state water-table level implied by each well's fitted state-space coefficients under long-term mean climate. Because its forcing is the long-term normal rather than a specific observation window, it is climate-window-independent and computable from a shorter record than the five-year mean spring level (MSL5) requires. Section 3.7.6 of the main report defines the index and its uncertainty; Section 4.8.6 reports its calibration onto the MSL5 scale and its cross-validation against the Ellenberg-F vegetation data. This note carries the full per-well reconstruction that Section 4.8.6 refers to.

The calibration is scoped to the open-dune network (clusters C1–C3, reference and extended wells). Of the 64 open-dune wells carrying an index, 62 also carry an observed MSL5 and set the calibration; the remaining two have no valid five-year spring window and are reconstructed only. The 20 forest wells (C4, C5) are reconstructed but held out of scope: their coefficients are the least constrained on the site, and the reconstruction there is too coarse to place a well across a Curreli et al. (2013) slack threshold. All 84 wells are listed, with the scoping carried explicitly in the Status column, so that the boundary of the calibration is visible rather than implied. CEH13 and CEH14 do not appear: the drainage coefficient is the denominator of the equilibrium expression and is near zero or negative at both, leaving the index undefined.

Standard errors propagate uncertainty in the fitted coefficients through the equilibrium expression. The variant tabulated here propagates the drainage coefficient alone, which is the dominant term because the equilibrium displacement is inversely proportional to it; a first-order propagation over all three coefficients is carried alongside it in the source CSV. The propagation is anchored on the equilibrium displacement rather than on the datum-referenced index value: subtracting the constant drainage datum shifts the value but not its uncertainty, and anchoring on the shifted value understates the error, severely so at wells whose equilibrium level sits close to the datum. These standard errors are substantially larger than the sampling error on a five-year MSL5 mean at every well, which is the basis for treating the index as a complement to MSL5 rather than a replacement for it.

Table S7.1. Equilibrium wetness index and MSL5 reconstruction, per well ($n = 84$). Levels are metres below ground surface, negative below the surface. The index is reconstructed from the open-dune calibration $MSL5 = +0.237 + 0.929 \cdot EWI$ ($r = 0.98$, $RMSE = 62$ mm), fitted on the 62 in-scope wells carrying an observed five-year mean spring level. A further 2 open-dune well(s) carry an index but no valid five-year spring window and are reconstructed only; the 20 C4/C5 forest wells are reconstructed but held out of scope, their coefficients being the least constrained on the site. Status reads: 'Calibration', an in-scope well that entered the fit; 'Reconstructed', in scope but with no observed spring level; 'Out of scope', a forest well outside the calibration's scope. Standard errors on the index propagate the drainage coefficient alone, the dominant term; the full three-coefficient variant is carried in 26_equilibrium_wetness_index_per_well.csv. Source: 26_equilibrium_wetness_index_per_well.csv, 26_ewi_msl5_comparison.csv.

Well	Network	Cluster	β_3 (mth^{-1})	EWI (m bg)	SE (mm)	MSL5 obs	MSL5 recon	Resid. (mm)	Status
CEH11	Reference	C1 (Lake Edge)	0.0907	-0.628	328	-0.400	-0.346	54	Calibration
CEH23	Reference	C1 (Lake Edge)	0.0884	-0.530	368	-0.278	-0.255	24	Calibration
CEH25	Reference	C1 (Lake Edge)	0.1333	-0.522	304	-0.326	-0.248	79	Calibration
CEH26	Reference	C1 (Lake Edge)	0.0865	-0.493	364	-0.268	-0.221	47	Calibration
CEH27	Reference	C1 (Lake Edge)	0.0982	-0.459	319	-0.212	-0.189	23	Calibration
CEH5	Reference	C1 (Lake Edge)	0.0972	-0.514	320	-0.263	-0.240	24	Calibration
CEH6	Reference	C1 (Lake Edge)	0.1175	-0.381	288	-0.238	-0.117	121	Calibration
P2	Extended	C1 (Lake Edge)	0.0990	-0.503	364	-0.326	-0.230	96	Calibration
CEH10	Reference	C2 (Dune)	0.1039	-1.374	347	-1.070	-1.039	31	Calibration
CEH24	Reference	C2 (Dune)	0.0737	-0.432	476	-0.117	-0.164	-47	Calibration
CEH28	Reference	C2 (Dune)	0.0712	-0.603	491	-0.256	-0.323	-67	Calibration
D10	Reference	C2 (Dune)	0.0577	-0.461	492	-0.174	-0.191	-17	Calibration
D15	Reference	C2 (Dune)	0.0611	-0.405	460	-0.071	-0.139	-68	Calibration
D17	Reference	C2 (Dune)	0.0708	-0.427	517	-0.163	-0.160	3	Calibration
D38	Reference	C2 (Dune)	0.1085	-0.741	285	-0.433	-0.451	-18	Calibration
D41	Reference	C2 (Dune)	0.0656	-0.480	467	-0.207	-0.209	-2	Calibration
D43	Reference	C2 (Dune)	0.0754	-0.659	343	-0.363	-0.375	-12	Calibration
D44	Reference	C2 (Dune)	0.0877	-0.741	417	-0.440	-0.451	-11	Calibration
D5	Reference	C2 (Dune)	0.0679	-0.592	473	-0.279	-0.313	-34	Calibration
D6	Reference	C2 (Dune)	0.0651	-0.578	457	-0.265	-0.299	-34	Calibration
D7	Reference	C2 (Dune)	0.0765	-0.470	449	-0.261	-0.199	62	Calibration
D8	Reference	C2 (Dune)	0.0553	-0.436	536	-0.159	-0.168	-9	Calibration
D9	Reference	C2 (Dune)	0.0621	-0.567	463	-0.257	-0.290	-33	Calibration
L7	Reference	C2 (Dune)	0.0841	-1.598	250	-1.300	-1.247	53	Calibration
NW3	Reference	C2 (Dune)	0.0615	-0.594	551	-0.298	-0.314	-16	Calibration
NW4	Reference	C2 (Dune)	0.0540	-0.443	588	-0.188	-0.174	14	Calibration
NW4B	Reference	C2 (Dune)	0.0593	-0.472	569	-0.211	-0.201	10	Calibration
T41A	Reference	C2 (Dune)	0.0940	-0.487	405	-0.174	-0.215	-41	Calibration
T41B	Reference	C2 (Dune)	0.0649	-0.399	483	-0.159	-0.133	26	Calibration
T41C	Reference	C2 (Dune)	0.0698	-0.436	412	-0.177	-0.167	9	Calibration
T41D	Reference	C2 (Dune)	0.0669	-0.662	444	-0.375	-0.377	-3	Calibration
WMC1	Reference	C2 (Dune)	0.0555	-0.599	548	-0.309	-0.319	-10	Calibration
CEH12	Extended	C2 (Dune)	0.0849	-1.188	383		-0.866		Reconstructed
CEH22	Extended	C2 (Dune)	0.0563	-0.707	237	-0.557	-0.420	137	Calibration
CEH35	Extended	C2 (Dune)	0.0658	-0.800	389	-0.652	-0.506	147	Calibration
CEH37	Extended	C2 (Dune)	0.0780	-0.542	349	-0.119	-0.266	-147	Calibration
CEH7	Extended	C2 (Dune)	0.0478	-0.687	661	-0.266	-0.401	-135	Calibration
CEH8	Extended	C2 (Dune)	0.0322	-0.772	510	-0.363	-0.480	-117	Calibration
PW	Extended	C2 (Dune)	0.0702	-0.882	441	-0.609	-0.582	27	Calibration
CEH1	Reference	C3 (Western Residual)	0.0653	-0.713	417	-0.277	-0.425	-148	Calibration
CEH18	Reference	C3 (Western Residual)	0.0543	-1.008	531	-0.695	-0.699	-4	Calibration
CEH21	Reference	C3 (Western Residual)	0.0414	-1.151	428	-0.923	-0.832	91	Calibration
CEH36	Reference	C3 (Western Residual)	0.0454	-0.745	493	-0.485	-0.454	31	Calibration
CEH39	Reference	C3 (Western Residual)	0.0307	-0.483	803	-0.253	-0.212	41	Calibration
CEH4	Reference	C3 (Western Residual)	0.0667	-0.859	309	-0.605	-0.560	45	Calibration
CEH40	Reference	C3 (Western Residual)	0.0345	-0.623	867	-0.382	-0.341	41	Calibration
CEH41	Reference	C3 (Western Residual)	0.0344	-0.478	848	-0.231	-0.207	24	Calibration
CEH42	Reference	C3 (Western Residual)	0.0506	-0.511	536	-0.201	-0.238	-36	Calibration
CEH9	Reference	C3 (Western Residual)	0.0330	-0.712	736	-0.451	-0.424	26	Calibration
D25	Reference	C3 (Western Residual)	0.0636	-0.644	422	-0.260	-0.361	-100	Calibration
NW1	Reference	C3 (Western Residual)	0.1013	-1.637	306	-1.307	-1.283	23	Calibration
NW11	Reference	C3 (Western Residual)	0.0747	-0.882	360	-0.521	-0.582	-60	Calibration
NW13	Reference	C3 (Western Residual)	0.0868	-0.803	317	-0.481	-0.509	-28	Calibration
NW2	Reference	C3 (Western Residual)	0.0789	-0.831	408	-0.486	-0.535	-49	Calibration

Well	Network	Cluster	β_3 (mth ⁻¹)	EWI (m bg)	SE (mm)	MSL5 obs	MSL5 recon	Resid. (mm)	Status
NW5	Reference	C3 (Western Residual)	0.0498	-0.735	590	-0.407	-0.446	-38	Calibration
NW6	Reference	C3 (Western Residual)	0.0517	-0.628	550	-0.301	-0.346	-44	Calibration
NW7	Reference	C3 (Western Residual)	0.0409	-1.422	603	-1.030	-1.084	-54	Calibration
WMC2	Reference	C3 (Western Residual)	0.0469	-0.623	544	-0.347	-0.342	5	Calibration
WMC3	Reference	C3 (Western Residual)	0.0316	-1.350	776	-0.976	-1.016	-40	Calibration
WMC4	Reference	C3 (Western Residual)	0.0871	-0.836	304	-0.518	-0.539	-21	Calibration
CEH38	Extended	C3 (Western Residual)	0.0555	-0.779	433	-0.599	-0.486	113	Calibration
NW8B	Extended	C3 (Western Residual)	0.0594	-1.592	342	-1.205	-1.242	-36	Calibration
P1	Extended	C3 (Western Residual)	0.0431	-0.319	868		-0.059		Reconstructed
PE	Extended	C3 (Western Residual)	0.0506	-0.732	550	-0.495	-0.443	52	Calibration
CEH2	Reference	C4 (Main Forest)	0.0176	-1.171	1644	-0.846	-0.851	-5	Out of scope
CEH20	Reference	C4 (Main Forest)	0.0278	-1.297	883	-0.927	-0.967	-40	Out of scope
CEH30	Reference	C4 (Main Forest)	0.0230	-1.664	984	-1.424	-1.308	116	Out of scope
CEH32	Reference	C4 (Main Forest)	0.0120	-1.435	2352	-1.336	-1.096	240	Out of scope
CEH33	Reference	C4 (Main Forest)	0.0190	-1.619	1265	-1.409	-1.267	143	Out of scope
CEH34	Reference	C4 (Main Forest)	0.0077	-0.381	3338	-0.484	-0.117	367	Out of scope
NW10	Reference	C4 (Main Forest)	0.0413	-1.020	626	-0.681	-0.710	-29	Out of scope
CEH15	Extended	C4 (Main Forest)	0.0361	-1.357	527	-0.810	-1.023	-213	Out of scope
FE1	Extended	C4 (Main Forest)	0.0141	-0.863	1499	-0.433	-0.565	-132	Out of scope
LIS1	Extended	C4 (Main Forest)	0.0257	-0.934	1166	-0.413	-0.630	-217	Out of scope
CEH16	Reference	C5 (Coastal Forest)	0.0364	-0.974	557	-0.715	-0.667	48	Out of scope
CEH17	Reference	C5 (Coastal Forest)	0.0546	-2.204	343	-1.961	-1.809	152	Out of scope
CEH19	Reference	C5 (Coastal Forest)	0.0420	-1.051	424	-0.831	-0.739	92	Out of scope
CEH31	Reference	C5 (Coastal Forest)	0.0385	-1.610	669	-1.381	-1.258	123	Out of scope
NW9	Reference	C5 (Coastal Forest)	0.0489	-0.933	434	-0.676	-0.629	47	Out of scope
CEH3	Extended	C5 (Coastal Forest)	0.0554	-1.170	262	-1.081	-0.849	231	Out of scope
FE2	Extended	C5 (Coastal Forest)	0.0455	-1.708	390	-1.341	-1.349	-9	Out of scope
FE3	Extended	C5 (Coastal Forest)	0.0603	-2.032	340	-1.726	-1.650	76	Out of scope
FE4	Extended	C5 (Coastal Forest)	0.0475	-1.621	486	-1.253	-1.268	-16	Out of scope
NW8	Extended	C5 (Coastal Forest)	0.0516	-1.484	361	-1.401	-1.141	259	Out of scope

S7.2 Reproducibility

The equilibrium wetness index, its calibration onto the MSL5 scale, and this table are computed by `26_van_willegen_msl.py`. Reference-tier coefficients are read from `03_master_data.csv` (Script 03) and extended-tier coefficients are fitted in the same script using the shared state-space routine in `utils/model_utils.py`. The per-well index and its standard errors are exported to `26_equilibrium_wetness_index_per_well.csv`, the observed-versus-reconstructed comparison to `26_ewi_msl5_comparison.csv`, and the calibration constants to `26_report_numbers.csv`. The table above is emitted directly from those outputs as `26_table_s7_1_ewi_per_well.csv` rather than transcribed, so it cannot drift from them.

Supplementary Note S8: Spring-Mean (MAM) Seasonal Robustness Analysis

S8.1 Purpose

The site's ecologically framed BACI and coastal-gradient analyses use the annual summer minimum (Jun–Sep) as the response metric — the drought-stress quantity that determines dune-slack habitat viability. This note tests whether that choice drives the conclusions, by re-running the seasonal analyses on the annual spring mean (Mar–May, the season the Curreli ecological thresholds are defined on) as a second metric through identical code paths. Spring is added, never substituted: the summer minimum remains the headline. The spring mean carries the same N as the summer minimum (one value per well-year) and, being a three-month mean rather than an extreme-order statistic, is the less noisy of the two. The spring window and its strict 3-of-3 completeness rule are the config constants `MSL_SPRING_MONTHS = (3, 4, 5)` and `MSL_MIN_MONTHS_PER_SPRING = 3`, shared with the van Willegen MSL analyses (Methods Supplement F.4).

S8.2 Scraping Intervention (Script 09c)

At the scraped well CEH36, the spring mean shifts post-scraping by +117 mm against the paired CEH4 control ($p = 0.003$) and +46 mm against the climate centroid ($p = 0.54$, not significant), versus +195 mm and +161 mm respectively for the summer minimum. Scraping raises the spring level too, by less than it raises the summer minimum, and the two controls — which agreed in summer — disagree in spring because every non-impact well also shifts negative against the climate centroid, the signature of the regional spring centroid itself having risen rather than of the impact wells falling. With $n_{\text{pre}} = 4$ years these p -values are fragile.

S8.3 Clearfell BACI (Scripts 10d, 10l)

The clearfell tier-mean spring shifts against the Forest control (Script 10d) are Impact –15, Edge –101, Forest-control –9 and Climate-control –78 mm, none individually significant. The four-zone spring model (Script 10l; 14-year panel, $R^2 = 0.57$) returns the differential felling steps in Table S8.1. The Edge step (–46 mm, $p = 0.011$) is the only nominally significant spring BACI result and is not robust: it is distributed across all four Edge wells (it survives dropping any one, –36 to –58 mm) but is leveraged by a single pre-felling year — dropping 2011 collapses it to –11 mm ($p = 0.36$) — and is fitted with cluster-robust standard errors on very few clusters. It is reported for completeness, not as a felling effect. In summer the felling signal sits at the Impact zone; the spring branch does not reproduce a significant Impact step, consistent with the felling effect being a growing-season phenomenon. The spring panel spans 14 years (2011 + 2013–2025) against the summer panel's 13, because WMC3's Mar–May record is near-complete and 2019 — unusable in summer — is fully measured in spring, even under the stricter 3-of-3 rule.

Table S8.1. Four-zone differential felling step relative to the Forest control zone, spring mean versus summer minimum (Scripts 10l_06, 10l_01). Steps in mm; C3/Warren is the second control zone, expected near zero.

Zone	Spring step $\pm$ SE (mm)	Spring p	Summer step $\pm$ SE (mm)	Summer p
Impact	–14.7 $\pm$ 11.4	0.196	–7.3 $\pm$ 7.9	0.358
Edge	–46.4 $\pm$ 18.2	0.011	–28.6 $\pm$ 20.5	0.162

C3/Warren	-9.7 ± 11.7	0.407	$+12.0 \pm 10.6$	0.257
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S8.4 Cluster Trends and the Coastal Gradient (Scripts 14, 25)

The spring-mean cluster-centroid trend (Script 14) is flat for every cluster except C5 Coastal Forest (-0.038 m yr^{-1} , $p = 0.020$); C1–C4 lie between -0.001 and $+0.007 \text{ m yr}^{-1}$, none significant. The contrast with the summer minimum is the substantive result (Table S8.2, Figure S8.1): summer minima decline across every cluster (-9 to -36 mm yr^{-1}), but spring means are essentially stable inland — the inland summer-minimum decline is a growing-season phenomenon, not a fall in the baseline water table. C5 is the exception: it declines in both seasons (summer -35.9 , spring -37.9 mm yr^{-1}), and applying the all-season coastal-retreat gradient (Script 25, Methods Supplement S.15) to the spring metric attributes about 41 % of C5's spring decline to coastal retreat (predicted gradient -15.4 mm yr^{-1} on the constant -6.35 mm yr^{-1} climate background, residual -16.1 mm yr^{-1}). This is direct corroboration that C5's decline is a year-round coastal-boundary effect rather than a drought-season artefact.

The full-panel season $\times \delta(d) \cdot t$ interaction test (Methods Supplement S.15) confirms this at the gradient level: on the forest-free panel (11,777 observations, one model) the spring modulation of the gradient drift rate is $\gamma = +0.058 \pm 0.048$ ($t = 1.20$, $p = 0.23$) for the linear-capped form and $+0.059 \pm 0.048$ ($p = 0.22$) for the exponential. The coastal-retreat drift is about 6 % steeper in spring as a point estimate but statistically indistinguishable from season-independent, so the same all-season gradient applies to both seasons. A MAM-only refit of the gradient panel (a sensitivity block in 25_01_panel_fit_parameters.csv) loses power as expected — the coast-edge anomaly δ_0 standard error roughly doubles (1.92 to 3.34 mm yr^{-1}) — but leaves the point estimates essentially unchanged ($\delta_0 -31.0 \text{ mm yr}^{-1}$, $L = 894 \text{ m}$).

Table S8.2. Observed cluster-centroid slope, summer minimum versus spring mean, with the spring-mean decomposition under the all-season forest-free linear-capped gradient (Scripts 25_08, 25_03_cluster_partition_spring). Slopes in mm yr^{-1} ; the climate background is a constant -6.35 mm yr^{-1} .

Cluster	Summer obs	Spring obs	Spring coastal gradient	Spring residual
C1 (Lake Edge)	-10.9	+6.5	0.0	+12.9
C2 (Dune)	-11.2	+0.3	0.0	+6.7
C3 (Western Residual)	-9.4	-1.0	-2.2	+7.6
C4 (Main Forest)	-17.8	-1.2	0.0	+5.2
C5 (Coastal Forest)	-35.9	-37.9	-15.4	-16.1

Figure S8.1 (25_08_spring_vs_summer_comparison.png) shows the observed centroid slope by cluster, summer minimum versus spring mean, with the all-season gradient marked. Figure S8.2 (14_climate_trajectory_spring.png) shows the observed spring-mean trajectories by cluster with per-cluster OLS trends.

S8.5 Conclusion

The spring-mean analysis confirms the summer-minimum conclusions rather than revising them. Scraping raises spring levels as it does summer minima; the qualitative clearfell-BACI picture does not depend on the choice of summer minimum over spring mean; the inland summer-minimum decline is growing-season specific (spring flat); and — via the null interaction test and the year-round C5 signal — the coastal-retreat gradient is season-

independent. The one nominally significant spring-only result, the Edge four-zone step, is not robust to a single pre-felling year and is not interpreted as a felling effect.

S8.6 Reproducibility

Scripts 09c (v1.5.0), 10d (v1.7.0), 10l (v1.2.0), 14 (v1.4.x), 25 (v1.5.x). Every spring output is additive and the committed summer outputs reproduce byte-identically. The spring season and completeness rule are `config.MSL_SPRING_MONTHS` and `config.MSL_MIN_MONTHS_PER_SPRING`; the interaction test and MAM sensitivity are described in Methods Supplement S.15.